MTH6134 Statistical Modelling II

Group Project Brief

Midland Premier League 2023/24 – Meridian Analytics

Context

Meridian Analytics has been commissioned by Ashford United FC to produce a statistical analysis of data from the 2023/24 Midland Premier League season. Your group will work as a team of statistical analysts and produce a written report. All data are simulated; the league, clubs, and players are fictional.

The Data

The project uses the two datasets released at the start of term, `league_players.csv` and `ashford_matches.csv`. They are described in full in the Data Brief document.

No prior knowledge of football is needed. The datasets and the Data Brief, which defines every variable, have been available from the start of term, and the Data Brief gives week-by-week suggestions for models to fit, with the chance to ask me and the teaching assistants in the PC sessions throughout. The task is to explore relationships between the variables, and the interpretation is not expected to go beyond that into football-specific insights.

The Task

Your group is asked to identify three questions that can be answered from these data and to present a statistical analysis of each. Deciding which three questions are most worth presenting to the club, and justifying your choice, is part of the task.

Choose questions that give you sufficient scope to demonstrate your statistical understanding and analytical skills. A well-executed analysis of a very simple or limited question may not provide sufficient evidence to achieve the highest marks.

For each of your three analyses, choose an appropriate statistical model and justify the choice, carry out model selection, assess model fit, and interpret the results with appropriate uncertainty quantification.

Working as a Group

You have been exploring these data in the PC sessions since the start of term, following the suggestions in the Data Brief document, and keeping your work in a shared workspace as described in the Working as a Group document. The task above draws together the strands you have been building week by week.

Continue to keep everything in your shared workspace: all code including rough exploratory work, the plots and outputs you produce, notes on the analytical decisions your group makes and why (your decisions log), and a running record of who did what in the corresponding doc. This record is the evidence used to set each member's contribution factor (see Assessment); it also provides evidence of how your analysis developed. The workspace itself is not marked, but it is used to confirm that the report is the group's own work and to set each member's contribution factor.

The Group Report

The report is a professional analytical report written for an informed but non-specialist audience. It must be no more than **four pages** of narrative, followed by an R code appendix (does not count towards the page limit) containing the complete code used to produce every result and figure in the report and a link to your shared workspace. Your code should run and reproduce the reported results; reproducibility is marked under presentation.

The report should include an executive summary (no more than half a page) stating the three main findings, and for each analysis: a description of the data explored, the modelling approach chosen and why, the results with appropriate uncertainty quantification, and a brief diagnostic assessment of model fit. A short conclusions section should note the main findings and their limitations.

The Individual Report

Alongside the group report, each member submits an individual piece of work. It has three parts, with the word limits shown.

1. Individual analysis (500 words, plus one figure or table of your choice and an R code appendix that does not count towards the word limit).

You are given two further variables, in the dataset `ashford_matches_individual.csv`, that your group did not have: `shots_on_target` (shots on target by Ashford United in a match) and `possession` (percentage possession by Ashford United in a match). Using one or both of these, carry out an analysis of your own. You might, for example, extend one of your group's analyses to include a new variable, fit a model your group did not consider, or answer a new question, whatever you judge sensible. Choose an appropriate model, fit it, check it, and interpret the result. Compare what you did with your group's work where relevant, and discuss the advantages and disadvantages of using these variables.

2. Reflection on feedback (100 words).

Reflect on how the feedback you received on your week 7 individual report, and the experience of giving feedback to a peer, influenced both your group work and your individual analysis here.

3. Reflection on AI use (100 words).

This project is Green (Fully Permitted and Encouraged) under the QMUL AI Traffic Light System. Reflect critically on your use of AI: what you used it for and where it helped, where it was wrong or misleading and how you noticed, and what statistical judgement you had to apply yourself that it could not. If you did not use AI, reflect on that choice and what it meant for how you worked.

Assessment

This assessment contributes **60%** of your module mark. Within this, the group report contributes 40 percentage points and the individual report contributes 20 percentage points. Each member's share of the 40 percentage points from the group report is scaled by an individual contribution factor, described below.

Group report (marked out of 100)

Component	Marks
Choice of questions	15
Exploratory analysis	15
Model choice, selection and fit	20
Interpretation	30
Presentation and code	20
Total	**100**

Contribution factor

Each member's share of the group report mark is scaled by a contribution factor. The default is 1, and every member keeps 1 unless your Group Contribution Summary and the workspace records behind it show that someone did less than their share. These are the only evidence used to set it, so a member who contributed fairly is never marked down. The levels are 1 (a fair share of the work), 0.75 (less than a fair share), 0.5 (substantially less than their share), 0.25 (very little), and 0 (no evidence of any contribution). A member on 0.75 keeps 75% of the group report marks, one on 0.5 keeps half, and so on. A factor of 0 would follow only clear evidence of no contribution.

Individual report (marked out of 100)

Part	Marks
Individual analysis	50
Reflection on feedback	25
Reflection on AI use	25
Total	**100**

AI Use, Collaboration, and Academic Integrity

This project is classified as **Green** (Fully Permitted and Encouraged) under the QMUL AI Traffic Light System. You may use AI tools, and their use is encouraged. Your individual report must be your own work and must be written in your own words.

The group report is the collective work of your group. You are expected to divide the analysis among yourselves and to work together on it, but the report and each individual report must be your group's and your own work. You may not share work with, or use work from, other groups. Your individual report must be written independently. Standard QMUL academic integrity regulations apply to all submitted work.

Submission

The deadline is **23.59 on 6 January 2027**, the start of the exam period. Submission is via QM+; the full submission details will be confirmed on QM+.

Your final submission consists of

- **Group report**: 4 pages of text with the structure described on page 2 of this document (also see Writing tips document and feedback on week 7 report) plus R code appendix.
- **Group Contribution Summary**: a single page document that includes a short entry for each member that summarises what that member contributed to the decisions the group made and to the project you submitted, across the analyses, modelling, writing, code and coordination. Keep each entry specific and to what that person actually did. It is the one-page distillation of the record of who did what and the decisions log you keep in your shared workspace through the term, which stay in the workspace. I use it, with the workspace records behind it, to set each member's contribution factor. Reminder: the default contribution factor is 1 for all, unless the evidence in this document suggests otherwise.
- **Individual report**: submitted by each member individually. The 500-word individual analysis using the two additional variables (one figure or table, plus an R code appendix that does not contribute towards the word limit), the 100-word reflection on feedback, and the 100-word reflection on AI use, as described on page 2 of this document.